CS Fundamentals Revision Handbook

Prepo by Avneesh — OOP, DBMS, SQL, OS & Networks | Category: QUICK REVISION NOTES

1. OOP Principles

• Encapsulation: Bundling data and methods into a single unit; data hiding via access specifiers (private/protected/public).
• Abstraction: Hiding implementation details and exposing only essential interfaces (Interfaces & Abstract Classes).
• Inheritance: Reusability mechanism (IS-A relationship); Single, Multilevel, Hierarchical, Multiple (via interfaces).
• Polymorphism: Compile-time (Method Overloading) vs Run-time (Method Overriding with dynamic dispatch).

2. DBMS & SQL Core

• ACID Properties: Atomicity (All or nothing), Consistency (State preservation), Isolation (Concurrent safety), Durability (Persisted).
• Normalization: 1NF (Atomic attributes), 2NF (No partial functional dependency), 3NF (No transitive dependency), BCNF.
• SQL Joins: INNER (matching records), LEFT/RIGHT (all from left/right + matching), FULL OUTER, CROSS (Cartesian).
• Group By & Having: GROUP BY aggregates rows; HAVING filters aggregated groups (WHERE cannot filter aggregations).

3. Operating Systems & Computer Networks

• Deadlock Coffman Conditions: Mutual Exclusion, Hold & Wait, No Preemption, Circular Wait (All 4 must hold).
• Process vs Thread: Process has independent address space; threads share heap, code, data of parent process.
• OSI 7 Layers: Physical -> Data Link -> Network (IP) -> Transport (TCP/UDP) -> Session -> Presentation -> Application.
• TCP vs UDP: TCP is connection-oriented, reliable, with 3-way handshake (SYN, SYN-ACK, ACK); UDP is connectionless & fast.